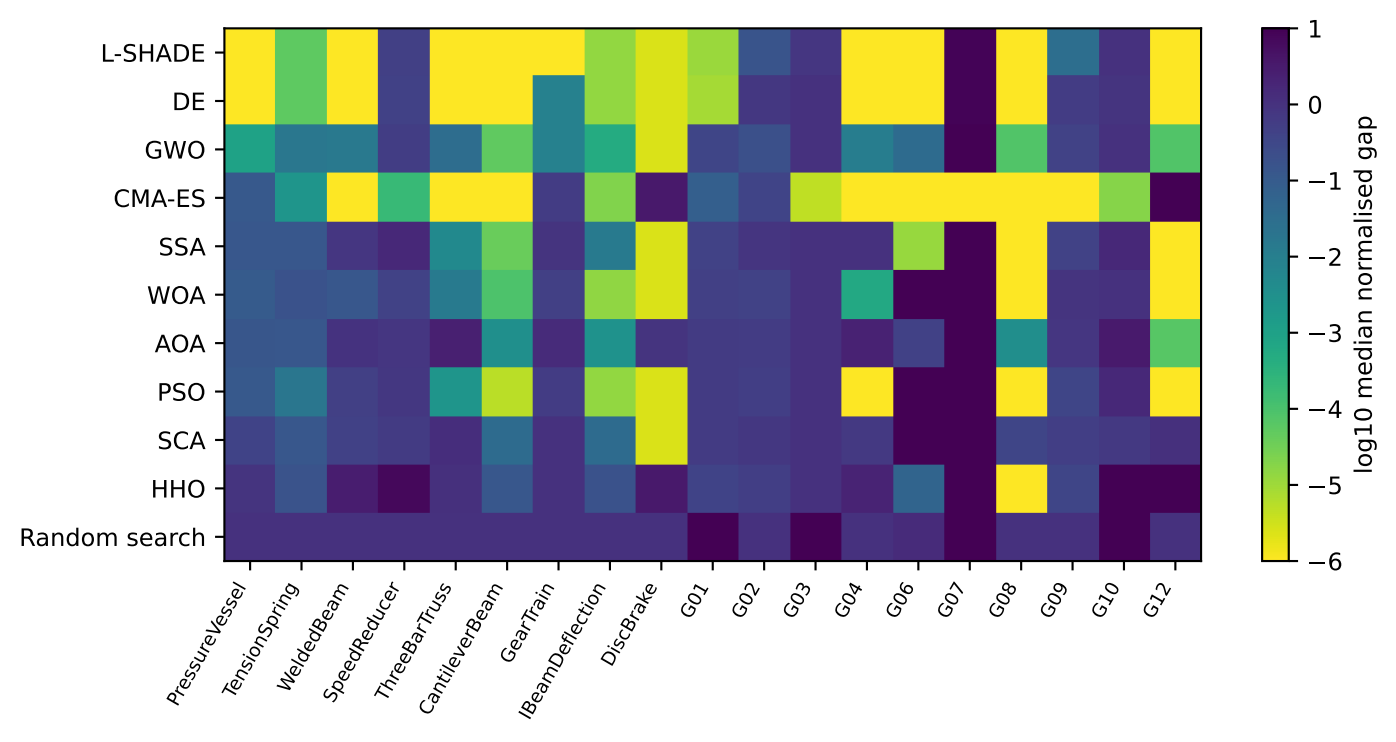
L-SHADE
DE
GWO
CMA-ES
SSA
WOA
AOA
PSO
SCA
HHO
Random search
PressureVessel
TensionSpring
WeldedBeam
SpeedReducer
ThreeBarTruss
CantileverBeam
GearTrain
IBeamDeflection
DiscBrake
G01
G02
G03
G04
G06
G07
G08
G09
G10
G12
log10 median normalised gap
1
0
−1
−2
−3
−4
−5
−6